

CyberForge AI

AI-Powered Ethical Hacking Workbench & Security Assessment Reporting Platform

CyberForge AI is a unified security-assessment workstation designed for authorized ethical hackers. It streamlines the workflow from defining the rules of engagement and executing scopes to AI-assisted analysis and professional PDF report generation.

Features

- **Strict Authorization Modeling:** Built-in scope enforcement ensures tools only execute against authorized targets.
- **Tool Integration Engine:** Asynchronously runs native tools like `nmap` and parses XML/JSON output directly into the database.
- **AI Copilot (Provider Abstraction):** Connects to OpenAI, Anthropic, or Azure OpenAI to correlate findings and explain complex vulnerabilities. *Never executes autonomous attacks.*
- **Evidence Management:** Upload screenshots and PCAPs; the system automatically generates SHA-256 hashes for a cryptographic audit trail.
- **Professional Reporting:** 1-click generation of corporate-styled PDF reports using WeasyPrint and Jinja2.

Installation & Setup

Option 1: Docker (Recommended)

```
# 1. Clone the repository
git clone [https://github.com/yourusername/cyberforge-ai.git](https://github.co
cd cyberforge-ai

# 2. Configure Environment
cp .env.example .env
# Edit .env with your AI Provider API Keys

# 3. Launch the Platform
docker-compose up -d --build
```

Access the dashboard at: `http://localhost:8000`

Option 2: Local Virtual Environment

```
python3 -m venv .venv
source .venv/bin/activate # Windows: .venv\Scripts\activate
pip install -r requirements.txt
```

```
# Ensure system dependencies are installed:  
# Ubuntu/Debian: sudo apt install nmap libpango-1.0-0 libcairo2  
  
python main.py
```

Legal & Ethical Disclaimer

This application is strictly designed for **Authorized Security Assessments**. The operator is solely responsible for ensuring explicit, written consent (Rules of Engagement) is obtained before executing any security tools against target infrastructure. CyberForge AI enforces scope programmatically but does not replace human legal responsibility.